[536709] Stochastic Processes

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2 Introduction to Stochastic Processes

The random fluctuation along the time often appears inside phenomena both in the nature and society. The motion of micro-particles from pollen on the water surface, the time-dependent deviation of the number of customers at the service desk in the bank are well-known examples of such a random fluctuation. Several factors affects each other in a complicated manner and consequently these random motion or fluctuation appears. To analyze the properties of such a fluctuation, stochastic process has been studied as a stochastic modeling.

2.1 Discrete-Time Stochastic Process

Imagine tossing a dice several times and write the number at n-th tossing by X_n, or observe the motion of particle randomly collide other molecules and write the position at n-th observation by X_n. Then X_n gives the sequence randomly fluctuating. We call such a randomly fluctuating sequence (X_n) *descrete-time stochastic process.*

We consider $\mathbb{Z}_+ = \{0, 1, 2, \cdots\}$ which is discrete time. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We define discrete **stochastic process** as a collection of random variables on measurable space. Through this section we consider S to be a countable set. In general let S be a topological space (or compact metric space) and $\mathcal{B} = \mathcal{B}(S)$ be a Borel set which is a σ-algebra generated by the family of open sets in S.

Definition 2.1 (Discrete-time stochastic process)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $(S, \mathcal{B})$ be a measurable space and $(X_n)_{n\in\mathbb{Z}_+}$ be a collection of random variables

$$X_n \colon \Omega \to S \tag{2.1}$$

Then $\mathbf{X}(\omega) = (X_n(\omega))_{n\in\mathbb{Z}_+}$ is called **stochastic process** and $(S, \mathcal{B})$ is called **state space**.

2.2 Example: Random Walk

We consider one typical example of stochastic processes. The random walk model is applicable to various fields, like Physics, Chemistry, Finance and so on.

Definition 2.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\xi_j)_{j=1,2,\cdots}$ be a collection of random variables with

$$\mathbb{P}(\xi_j = 1) = p \qquad \mathbb{P}(\xi_j = -1) = 1 - p =: q \qquad j = 1, 2, \cdots \tag{2.2}$$

where $0 < p < 1$. We define 1-dimensional **simple random walk** starting from $x \in \mathbb{Z}$ by

$$S_0 = x, \qquad S_n = x + \xi_1 + \xi_2 + \cdots + \xi_n \qquad n = 1, 2, \cdots \tag{2.3}$$

We denote by $S = \{S_n\}_{n\in\mathbb{Z}_+}$.

When the case $p = \frac{1}{2}$, the simple random walk is *symmetric*. The increments $S_{n+1} - S_n = \xi_{n+1}$ take value $\{+1, -1\}$ and are i.i.d. (independently identically distributed). When we fix $\omega \in \Omega$, the sequence $S(\omega) := \{S_n(\omega)\}_{n\in\mathbb{Z}_+}$ is called a **path** of random walk. We have several properties derived from the fact that random walk is the sum of i.i.d. random variables.

- Average:
$$\mathbb{E}[S_n] = x + \sum_{k=1}^{n} \mathbb{E}[\xi_k] = x + n(p - q) \tag{2.4}$$
- Variance:
$$\mathbb{V}\mathrm{ar}(S_n) = n(1 - (p - q)^2) \tag{2.5}$$
- Distribution:
$$\mathbb{P}(S_n = y) = \begin{cases} 0 & \text{if } y - x + n \text{ is odd} \\ \binom{n}{k} p^k q^{n-k} & \text{if } y - x + n = 2k \end{cases} \tag{2.6}$$
- Independent Increment:
$S_{n+m} - S_m = \xi_{m+1} + \cdots + \xi_{m+n}$ is independent of S_j $(j = 1, 2, \cdots, m)$. Moreover, $\{S_{n+m} - S_m\}_{n=0,1,\cdots}$ is also a simple random walk starting from the origin.

2.2.1 Mathematical Construction of Random Walk

Coin tossing Random walk model is frequently explained by coin-tossing. We first consider coin-tossing: the probability that head of coin appears is $p \in (0, 1)$ and tail appears with probability $q = 1 - p$. We define the realization of ϵ_k which takes values $\{+1, -1\}$ by

$$\{\text{Head appears in } k\text{-th trial}\} = \{\epsilon_k = 1\}, \qquad \{\text{Tail appears in } k\text{-th trial}\} = \{\epsilon_k = -1\}$$

Thus

$$P(\epsilon_1 = i_1, \epsilon_2 = i_2, \cdots, \epsilon_n = i_n) = \prod_{j=1}^{n} P(\epsilon_j = i_j) \tag{2.7}$$

Mathematical modeling Let P be a Lebesgue measure on $[0,1)$. We first separate the interval $[0,1)$ with proportion q and p, i.e.

$$J_{-1} \cup J_1 = [0,q) \cup [q,1) = [0,1)$$

Define the function ξ_1 on $[0,1)$ by

$$\xi_1(\omega) = \begin{cases} -1 & \omega \in J_{-1} \\ 1 & \omega \in J_1 \end{cases}$$

Next we again separate J_{-1} and J_1 with proportion q and p, i.e.

$$J_{-1,-1} \cup J_{-1,1} = [0,q^2) \cup [q^2,q) = J_{-1}$$

$$J_{1,-1} \cup J_{1,1} = [q,1-p^2) \cup [1-p^2,1) = J_1$$

We also define the function ξ_2 by

$$\xi_2(\omega) = \begin{cases} -1 & \omega \in J_{-1,-1} \\ 1 & \omega \in J_{-1,1} \end{cases}$$

Then we define $J_{i_1,i_2,\cdots,i_n}$ and the function ξ_n repeatedly. Then

$$\{\omega \in [0,1) \colon \xi_1(\omega) = i_1, \xi_2(\omega) = i_2, \cdots, \xi_n(\omega) = i_n\} = J_{i_1,i_2,\cdots,i_n} \tag{2.8}$$

which implies that

$$\begin{aligned} P(\xi_1 = i_1, \xi_2 = i_2, \cdots, \xi_n = i_n) &= \text{Lebesgue measure of } J_{i_1,i_2,\cdots,i_n} \\ &= p^k\, q^{n-k} \\ &= \prod_{j=1}^{n} P(\xi_j = i_j) \end{aligned} \tag{2.9}$$

where k is the number of 1 in $i_1, i_2, \cdots, i_n$.

Summary Therefore,

the functions $\{\xi_1, \xi_2, \cdots, \xi_n\}$ is a mathematical model of coin-tossing $\{\epsilon_1, \epsilon_2, \cdots, \epsilon_n\}$.

2.3 Conditional Expectation

We define conditional probability and expectation in Note_1. In this section, we generalize the conditional probability and conditional expectation with respect to a σ-algebra $\mathcal{G} \subset \mathcal{F}$.

Definition 2.3 (Conditional probability)**.** Let Q be a function defined on $\mathcal{F} \times \Omega$. Then we call Q **conditional probability** (to be more precise, **regular conditional probability**) if Q satisfies the following conditions:

(i) for all $\Lambda \in \mathcal{F}$, $Q(\Lambda, \cdot)$ is a $\mathcal{G}$-measurable random variable.

(ii) for almost all $\omega \in \Omega$, $Q(\cdot, \omega)$ is a probability on Ω.

(iii) For any $\Xi \in \mathcal{G}$ and $\Lambda \in \mathcal{F}$,

$$\mathbb{P}(\Lambda \cap \Xi) = \mathbb{E}[Q(\Lambda, \cdot)\, \mathbb{1}_\Xi]. \tag{2.10}$$

We denote by $\mathbb{P}(\Lambda \mid \mathcal{G})(\omega) := Q(\Lambda, \omega)$ the conditional probability of Λ with respect to $\mathcal{G}$. Especially, for $\mathcal{G} = \sigma(X_1, X_2, \cdots, X_n)$, we sometimes write $\mathbb{P}(\Lambda \mid X_1, X_2, \cdots, X_n$.

Definition 2.4 (Conditional expectation)**.** Let X be an integrable random variable. Then the expectation by $\mathbb{P}(\cdot \mid \mathcal{G})$ is called **conditional expectation** of X with respect to $\mathcal{G}$ and denoted by $\mathbb{E}[X \mid \mathcal{G}]$.

The condition that a random variable X and σ-algebra $\mathcal{G}$ is independent is

$$\mathbb{P}(X \in A \mid \mathcal{G}) = \mathbb{P}(X \in A), \tag{2.11}$$

which also implies if X and $\mathbb{1}_\Xi$ for any $\Xi \in \mathcal{G}$ are independent each other, then X and $\mathcal{G}$ are independent.

Theorem 2.5. *The conditional expectation of X w.r.t. $\mathcal{G}$ is characterized uniquely as a random variable ξ satisfying*

(i) ξ is $\mathcal{G}$-measurable.

(ii) for any $\Xi \in \mathcal{G}$, $\mathbb{E}[X\mathbb{1}_\Xi] = \mathbb{E}[\xi\mathbb{1}_\Xi]$

Proof. It is trivial from the definition that $\mathbb{E}[X \mid \mathcal{G}]$ satisfies (i) and (ii). So we prove the uniqueness. Let ξ and η satisfy (i), (ii) and $\Xi = \{\omega \in \Omega \colon \xi(\omega) - \eta(\omega) > 0\}$. Assume that $\mathbb{P}(\Xi) > 0$. Then since $\Xi \in \mathcal{G}$,

$$0 < \mathbb{E}[(\xi - \eta)\,\mathbb{1}_\Xi] = \mathbb{E}[X\mathbb{1}_\Xi] - \mathbb{E}[X\mathbb{1}_\Xi] = 0, \tag{2.12}$$

which yields a contradiction. Thus $\mathbb{P}(\Xi) = 0$. Similarly, we can show that $\mathbb{P}(\xi - \eta < 0) = 0$. Therefore, we obtain $\xi = \eta$ almost surely. ■

Remark 2.6. For any $A \in \mathcal{G}$,

$$\mathbb{E}[X\mathbb{1}_A] = \int_A X\, d\mathbb{P} =: \nu(A) \tag{2.13}$$

Since the measure ν is absolutely continuous with respect to $\mathbb{P}$, by Radon–Nikodym theorem, there exists a random variable ξ such that

$$\nu(A) = \int_A \xi\, d\mathbb{P} = \mathbb{E}[\xi\mathbb{1}_A], \tag{2.14}$$

which implies

$$\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[\xi\mathbb{1}_A] \tag{2.15}$$

□

Corollary 2.7. *Let Z be a $\mathcal{G}$-measurable random variable and $\xi = \mathbb{E}[X \mid \mathcal{G}]$. Then*

$$\mathbb{E}[XZ] = \mathbb{E}[\xi Z] \tag{2.16}$$

The following proposition gives properties of conditional expectation.

Proposition 2.8. (i) $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$

(ii) $\mathbb{E}\left[\sum_{k=1}^{n} a_k X_k \middle| \mathcal{G}\right] = \sum_{k=1}^{n} a_k \mathbb{E}[X_k \mid \mathcal{G}]$ *a.s.*

(iii) *if* $X_1 \leq X_2$ *a.s., then* $\mathbb{E}[X_1 \mid \mathcal{G}] \leq \mathbb{E}[X_2 \mid \mathcal{G}]$ *a.s.*

(iv) $|\mathbb{E}[X \mid \mathcal{G}]|^p \leq \mathbb{E}[X^p \mid \mathcal{G}]$ *a.s.* $(p \geq 1)$

(v) *if* Y *be* $\mathcal{G}$*-measurable, then* $\mathbb{E}[XY \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]$ *a.s.*

(vi) *if* $\mathcal{G}_1 \subset \mathcal{G}_2$*, then* $\mathbb{E}[X \mid \mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1]$

Proof. We skip the proof of (i)–(iv) for Question.

(v): For $\mathcal{G}$-measurable η, $Y\eta$ is also $\mathcal{G}$-measurable. Set $\xi = \mathbb{E}[X \mid \mathcal{G}]$. Thus

$$\mathbb{E}[XY\eta] = \mathbb{E}[X(Y\eta)] = \mathbb{E}[\xi(Y\eta)] = \mathbb{E}[(\xi Y)\eta] \tag{2.17}$$

Thus

$$\mathbb{E}[XY \mid \mathcal{G}] = \xi Y = Y\,\mathbb{E}[X \mid \mathcal{G}] \tag{2.18}$$

(vi): Set $\xi = \mathbb{E}[X \mid \mathcal{G}_2]$. Let η be a $\mathcal{G}_1$-measurable random variable. Since η is also $\mathcal{G}_2$-measurable,

$$\mathbb{E}[X\eta] = \mathbb{E}[\xi\eta] = \mathbb{E}[\mathbb{E}[\xi \mid \mathcal{G}_1]\,\eta] \tag{2.19}$$

Thus

$$\mathbb{E}[X \mid \mathcal{G}_1] = \mathbb{E}[\xi \mid \mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1] \tag{2.20}$$

■

2.4 Filtration

We construct the functions $\xi_1, \xi_2, \cdots$ as in the previous section. Then by (2.9), the functions $\xi_1, \xi_2, \cdots$ is a sequence of i.i.d. random variables with

$$P(\xi_j = 1) = p \qquad P(\xi_j = -1) = q \qquad j = 1, 2, \cdots$$

Therefore, $S_0 = 0$ and

$$S_n = \sum_{k=1}^{n} \xi_k$$

is a random walk starting from the origin. On $J_{i_1,i_2,\cdots,i_n}$

$$S_k(\omega) = i_1 + i_2 + \cdots + i_k \quad k \leq n$$

is deterministic constant, which means that $J_{i_1,i_2,\cdots,i_n}$ yields *information* of S_k $(k \leq n)$.

Definition 2.9. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If the σ-algebra $\mathcal{F}_n \subset \mathcal{F}$ satisfies $\mathcal{F}_n \subset \mathcal{F}_{n+1}$, then $\{\mathcal{F}_n\}_{n \in \mathbb{Z}_+}$ is called **filtration** on $(\Omega, \mathcal{F}, \mathbb{P})$ and $(\Omega, \mathcal{F}, \{\mathcal{F}_n\}, \mathbb{P})$ is called **filtered probability space**.

The σ-algebra $\mathcal{F}_n$ generated by $\{J_{i_1,i_2,\cdots,i_n} : i_1, i_2, \cdots, i_n = -1, 1\}$ is filtration since the σ-algebra $\mathcal{F}_n = \sigma(S_1, S_2, \cdots, S_n)$ satisfies $\mathcal{F}_n \subset \mathcal{F}_{n+1}$. Indeed, the filtration $\mathcal{F}_{n+1}$ gives information

$$J_{i_1,i_2,\cdots,i_n,-1} \quad \text{and} \quad J_{i_1,i_2,\cdots,i_n,1}$$

which is more detailed information than $J_{i_1,i_2,\cdots,i_n}$. It is quite reasonable that we can get more information when the time passed. More precise condition for filtration is $\mathcal{N} \subset \mathcal{F}_0 \subset \cdots \subset \mathcal{F}_n \subset \cdots$ where $\mathcal{N}$ is the family of all null sets with respect to $\mathbb{P}$.

Next we consider a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$. For a stochastic process the state at each step should be "measurable" with the information at that time, that is, X_n should be $\mathcal{F}_n$-measurable at each $n \in \mathbb{Z}_+$.

Definition 2.10. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A stochastic process $\{X_n\}_{n\in I}$ is said to be **adapted** with respect to the filtration $\{\mathcal{F}_n\}_{n\in I}$ if and only if X_n is $\mathcal{F}_n$-measurable for any $n \in I$.

For random walk S, when $\mathcal{F}_n \supset \sigma(S_1, S_2, \cdots, S_n)$, random walk S is $\{\mathcal{F}_n\}$-adapted. Without the concept $\{\mathcal{F}_n\}$-adapted, we might consider the meaningless stochastic models. (X_n) is $\{\mathcal{F}_n\}$-adapted means that the present random value X_n depends only on the information until time n, which implies the present status never depends on the future. If X_n is $\mathcal{F}_{n-1}$-measurable for each n, then the process (X_n) is called **predictable** ($\{\mathcal{F}_{n-1}\}$-adapted).

2.5 Stopping Time

Definition 2.11. Let $(\Omega, \mathcal{F}, \{\mathcal{F}_n\}_{n\geq 0}, \mathbb{P})$ be a filtered probability space and $\mathcal{N}$ be a null-set with respect to $\mathbb{P}$ including $\mathcal{F}_0$. The function τ taking values on $\{0, 1, 2, \cdots, \infty\}$ is called **$\mathcal{F}_n$-stopping time** if it satisfies

$$\{\omega \colon \tau(\omega) = n\} \in \mathcal{F}_n \qquad n = 0, 1, 2, \cdots \tag{2.21}$$

When $\mathbb{P}(\tau < \infty) = 1$, τ is called finite $\{\mathcal{F}_n\}$-stopping time.

The definition guarantees that we can judge whether we should stop now or not without knowing information from the future.

Example 2.12. Let $\{X_n\}_{n\in\mathbb{Z}_+}$ be a sequence variables and $\mathcal{F}_n = \sigma(X_0, X_1, \cdots, X_n)$. Then the hitting time at the Borel set A

$$\tau(\omega) = \min\{k \geq 0 \colon X_k(\omega) \in A\}$$

is $\{\mathcal{F}_n\}$-stopping time, where for $\min \varnothing\ \tau(\omega) = \infty$. Indeed,

$$\{\tau = n\} = \{X_0 \notin A, \cdots, X_{n-1} \notin A, X_n \in A\}$$

□

Example 2.13. The last hitting time at the origin Θ is not $\{\mathcal{F}_n\}$-stopping time. Indeed, $\{\Theta = n\} \notin \mathcal{F}_n$ since

$$\{\Theta = n\} = \{X_n = 0, X_{n+1} \neq 0, X_{n+2} \neq 0, \cdots\}$$

□

Theorem 2.14. *The following condition is equivalent to* (2.21)

$$\{\omega \colon \tau(\omega) \leq n\} \in \mathcal{F}_n \tag{2.22}$$

Proof. Since

$$\{\omega\colon \tau(\omega) \le n\} = \bigcup_{k=0}^{\infty} \{\omega\colon \tau(\omega) = k\},$$

(2.21) implies (2.22). Opposite also holds because

$$\{\omega\colon \tau(\omega) = n\} = \{\omega\colon \tau(\omega) \le n\} \cap \{\omega\colon \tau(\omega) \le n-1\}^c$$

■

Theorem 2.15. *If τ_k $(k = 1, 2, \cdots)$ is $\{\mathcal{F}_n\}$-stopping time, then*

$$\tau_1 \vee \tau_2, \quad \tau_1 \wedge \tau_2, \quad \tau_1 + \tau_2, \quad \sup_{n=1,2,\cdots} \tau_k, \quad \inf_{k=1,2,\cdots} \tau_k$$

are also $\{\mathcal{F}_n\}$-stopping times, where $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$

Proof.

- Since
$$\{\tau_1 \vee \tau_2 \le n\} = \{\tau_1 \le n\} \cap \{\tau_2 \le n\},$$
$\tau_1 \vee \tau_2$ is also $\{\mathcal{F}_n\}$-stopping time.
- Since
$$\{\tau_1 \wedge \tau_2 \le n\} = \{\tau_1 \le n\} \cup \{\tau_2 \le n\},$$
$\tau_1 \wedge \tau_2$ is also $\{\mathcal{F}_n\}$-stopping time.
- Since
$$\{\tau_1 + \tau_2 \le n\} = \bigcup_{k=0}^{n} \big(\{\tau_1 = k\} \cap \{\tau_2 = n-k\}\big),$$
$\tau_1 + \tau_2$ is also $\{\mathcal{F}_n\}$-stopping time.
- Since
$$\{\sup_k \tau_k \le n\} = \bigcap_{k=0}^{\infty} \{\tau_k \le n\},$$
$\sup_k \tau_k$ is also $\{\mathcal{F}_n\}$-stopping time.
- Since
$$\{\inf_k \tau_k \le n\} = \bigcup_{k=0}^{\infty} \{\tau_k \le n\},$$
$\inf_k \tau_k$ is also $\{\mathcal{F}_n\}$-stopping time.

■

Theorem 2.16. *Let τ be a stopping time. Define*

$$\mathcal{F}_\tau = \{\Lambda \in \mathcal{F}\colon \Lambda \cap \{\tau = n\} \in \mathcal{F}_n,\ n \in \mathbb{Z}_+\}$$

Then it is complete σ-algebra and τ is $\mathcal{F}_\tau$-measurable.

Proof Since

$$\Omega \cap \{\tau = n\} = \{\tau = n\} \in \mathcal{F}_n,$$

we can say that $\Omega \in \mathcal{F}_\tau$. If $\Lambda \in \mathcal{F}_\tau$, then, since

$$\Lambda^c \cap \{\tau = n\} = \{\tau = n\} \cap \big(\Lambda \cap \{\tau = n\}\big)^c \in \mathcal{F}_n,$$

similarly $\Lambda^c \in \mathcal{F}_\tau$. If Λ_k $(k = 1, 2, \cdots)$, then since

$$\left(\bigcup_{k=1}^{\infty} \Lambda_k\right) \cap \{\tau = n\} = \bigcup_{k=1}^{\infty} \big(\Lambda_k \cap \{\tau = n\}\big) \in \mathcal{F}_n$$

we have $\bigcup_{k=1}^{\infty} \Lambda_k \in \mathcal{F}_\tau$. Therefore, $\mathcal{F}_\tau$ is a σ-algebra.

Next, if $\Lambda \in \mathcal{N}$, then since

$$\Lambda \cap \{\tau = n\} \in \mathcal{N} \subset \mathcal{F}_n,$$

$\Lambda \in \mathcal{F}_\tau$ holds, which implies that $\mathcal{F}_\tau$ is complete.

For $\Lambda := \{\tau = k\}$, since

$$\Lambda \cap \{\tau = n\} = \begin{cases} \varnothing & (n \neq k) \\ \{\tau = n\} & (n = k) \end{cases}$$

We have $\Lambda \cap \{\tau = n\} \in \mathcal{F}_n$. Hence $\{\tau = n\} \in \mathcal{F}_\tau$ $(k = 0, 1, \cdots)$ and τ is $\mathcal{F}_\tau$-measurable. ■

Theorem 2.17. *If τ and $\tilde{\tau}$ are stopping times, then*

$$\mathcal{F}_{\tau \wedge \tilde{\tau}} = \mathcal{F}_\tau \cap \mathcal{F}_{\tilde{\tau}}$$

Then it is complete σ-algebra and τ is $\mathcal{F}_\tau$-measurable.

Proof. Let $\Lambda \in \mathcal{F}_{\tau \wedge \tilde{\tau}}$. Since $\tau \wedge \tilde{\tau}$ is also $\{\mathcal{F}_n\}$-stopping time,

$$\Lambda \cap \{\tau \wedge \tilde{\tau} \leq n\} \in \mathcal{F}_n.$$

On the other hand, since

$$\Lambda \cap \{\tau \leq n\} = \Lambda \cap \{\tau \wedge \tilde{\tau} \leq n\} \cap \{\tau \leq n\},$$

we have $\Lambda \cap \{\tau \leq n\} \in \mathcal{F}_n$ and then $\Lambda \in \mathcal{F}_\tau$. Similarly we obtain $\Lambda \in \mathcal{F}_{\tilde{\tau}}$. Therefore,

$$\mathcal{F}_{\tau \wedge \tilde{\tau}} = \mathcal{F}_\tau \cap \mathcal{F}_{\tilde{\tau}}$$

Let $\tilde{\Lambda} \in \mathcal{F}_\tau \cap \mathcal{F}_{\tilde{\tau}}$. Then, since

$$\tilde{\Lambda} \cap \{\tau \wedge \tilde{\tau} \leq n\} = \tilde{\Lambda} \cap \big(\{\tau \leq n\} \cap \{\tilde{\tau} \leq n\}\big) = \big(\tilde{\Lambda} \cap \{\tau \leq n\}\big) \cap \big(\tilde{\Lambda} \cap \{\tilde{\tau} \leq n\}\big),$$

we have $\tilde{\Lambda} \cap \{\tau \wedge \tilde{\tau} \leq n\} \in \mathcal{F}_n$. Therefore, $\tilde{\Lambda} \in \mathcal{F}_{\tau \wedge \tilde{\tau}}$. ■

Question 2.18 (Recurrence)**.** Let $a < 0 < b$ be integers, $S = \{S_n\}_{n \in \mathbb{Z}_+}$ be a symmetric simple random walk starting at 0.We define the stopping time τ as the first time when the symmetric SRW hits a or b, i.e.

$$\tau = \min\{n \geq 0 : S_n = a \text{ or } S_n = b\} \tag{2.23}$$

Then compute the probability $\mathbb{P}(S_\tau = a)$. Moreover, by taking $b \to \infty$, show that

$$\mathbb{P}(\tau_a < \infty) = 1 \tag{2.24}$$

where $\tau_a = \min\{n \geq 0 \colon S_n = a\}$

2.6 Martingale

In this section we consider the concept "fair" in the sense of probability theory, which is known as *martingale*.

Definition 2.19. The stochastic process $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ is called $\{\mathcal{F}_n\}$-**martingale** if for all $n \in \mathbb{Z}_+$,

(1). $\mathbf{X}(\omega)$ is $\{\mathcal{F}_n\}$-adapted: X_n are integrable for all $n \in \mathbb{Z}_+$

(2). $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ a.s.

If the condition (2) changes to

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n \qquad \text{a.s.}, \tag{2.25}$$

then the process is called $\{\mathcal{F}_n\}$-**submartingale**. For opposite inequality "$\leq$", the process is called $\{\mathcal{F}_n\}$-**supermartingale**. Martingale is sub and supermartingale.

Example 2.20. Let $(\xi_n)_{n\in\mathbb{Z}_+}$ be measurable, i.i.d. and $\mathbb{E}[\xi_k] = 0$, ξ_n be $\mathcal{F}_n$-measurable, and ξ_{n+1} is independent of $\mathcal{F}_n$. Then $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ with

$$X_n = \sum_{k=0}^{n} \xi_k \tag{2.26}$$

is a $\{\mathcal{F}_n\}$-adapted random walk. Moreover, $\mathbf{X}(\omega)$ is a $\{\mathcal{F}_n\}$-martingale.

Indeed,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[X_n + \xi_{n+1} \mid \mathcal{F}_n] = X_n + \mathbb{E}[\xi_{n+1}] = X_n \tag{2.27}$$

□

Question 2.21. Let $(X_n)_{n\in\mathbb{Z}_+}$ be a symmetric simple random walk starting at the origin, i.e.

$$X_0 = 0, \quad X_n = \sum_{k=1}^{n} \xi_k \tag{2.28}$$

with $\mathbb{P}(\xi_j = 1) = \mathbb{P}(\xi_j = -1) = \frac{1}{2}$. Then prove that the following are $\{\mathcal{F}_n\}$-martingales.

(1) $X_n^2 - n$

(2) $X_n^3 - 3nX_n$

Question 2.22. Let $(X_n)_{n\in\mathbb{Z}_+}$ be a asymmetric simple random walk starting at the origin, i.e.

$$X_0 = 0, \quad X_n = \sum_{k=1}^{n} \xi_k \tag{2.29}$$

with $\mathbb{P}(\xi_j = 1) = p$ and $\mathbb{P}(\xi_j = -1) = 1 - p$ ($p \neq \frac{1}{2}$). Then prove that $X_n - (2p-1)n$ is a $\{\mathcal{F}_n\}$-martingales.

2.6.1 Doob decomposition

The analysis of $\{\mathcal{F}_n\}$-submartingale is not so easy because of its increasing fluctuations. However, submartingale has a nice expression by a martingale and predictable parts.

Theorem 2.23 (Doob decomposition)**.** *Let $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ be an $\{\mathcal{F}_n\}$-submartingale. Then there exist unique $\{\mathcal{F}_n\}$-martingale $(M_n)_{n\in\mathbb{Z}_+}$ and predictable $(A_n)_{n\in\mathbb{Z}_+}$ satisfying*

(i) $X_n = M_n + A_n$ for $n \in \mathbb{Z}_+$

(ii) A_n is $\mathcal{F}_{n-1}$-measurable and $0 = A_0 \le A_1 \le \cdots$ a.s.

Proof. First we set

$$Y_{k+1} = X_{k+1} - \mathbb{E}[X_{k+1} \mid \mathcal{F}_k], \quad (k = 0, 1, 2, \cdots), \qquad Y_0 = X_0. \tag{2.30}$$

Then Y_k is $\mathcal{F}_k$-measurable and

$$\mathbb{E}[Y_{k+1} \mid \mathcal{F}_k] = 0, \qquad \text{a.s.} \tag{2.31}$$

Now we define

$$M_n := \sum_{k=0}^{n} Y_k. \tag{2.32}$$

Then M_n is $\mathcal{F}_n$-measurable and

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[M_n + Y_{n+1} \mid \mathcal{F}_n] = M_n, \qquad \text{a.s.,} \tag{2.33}$$

which means that M_n is an $\{\mathcal{F}_n\}$-martingale.

Next we set

$$Z_{k+1} = \mathbb{E}[X_{k+1} \mid \mathcal{F}_k] - X_k, \quad (k = 0, 1, \cdots), \qquad Z_0 = 0. \tag{2.34}$$

Then Z_k is $\mathcal{F}_{k-1}$-measurable and by submartingale property,

$$Z_k \ge 0 \qquad \text{a.s.} \tag{2.35}$$

Now we define

$$A_n := \sum_{k=0}^{n} Z_k. \tag{2.36}$$

Then A_n satisfies the condition (ii) and the condition (i) is also satisfied by the definitions of M_n and A_n.

In the end, we show the uniqueness of this decomposition. If we can take another decomposition $\widetilde{M}_n$ and $\widetilde{A}_n$, then by (i),

$$\widetilde{M}_n - M_n = \widetilde{A}_n - A_n \tag{2.37}$$

which yields that $\widetilde{M}_n - M_n$ is $\mathcal{F}_{n-1}$-measurable and has the martingale property. Therefore,

$$\widetilde{M}_n - M_n = \mathbb{E}[\widetilde{M}_n - M_n \mid \mathcal{F}_{n-1}] = \widetilde{M}_{n-1} - M_{n-1} \quad \text{a.s.} \tag{2.38}$$

Inductively we obtain $\widetilde{M}_n - M_n = \cdots = \widetilde{M}_0 - M_0 = 0$, which shows that $\widetilde{M}_n = M_n$. Hence we conclude that

$$\widetilde{M}_n = M_n, \quad \text{and} \quad \widetilde{A}_n = A_n \tag{2.39}$$

■

Question 2.24. Let $(S_n)_{n\in\mathbb{Z}_+}$ be a symmetric simple random walk. Then show the Doob decomposition for $X_n = S_n^2$.

Remark 2.25 (Significance of Doob Decomposition)**.** The Doob decomposition

$$X_n = M_n + A_n \tag{2.40}$$

provides a fundamental insight into the structure of submartingales by separating them into two distinct parts:

- **Martingale part** M_n represents the purely "random" and "fair" fluctuations (noise).
- **Predictable part** A_n represents the "systematic" increasing trend (drift).

This decomposition is a discrete-time analogue of the decomposition of *semimartingales* in continuous-time stochastic calculus. It allows us to apply powerful martingale inequalities and convergence theorems to a broader class of stochastic processes by focusing on the martingale component M_n. □

2.6.2 Optional sampling theorem

Theorem 2.26 (Optional sampling theorem)**.** *Let τ and σ be $\{\mathcal{F}_n\}$-stopping times satisfying*

$$0 \le \tau(\omega) \le N \qquad \omega \in \Omega \tag{2.41}$$

- *If $\mathbf{X} = (X_n)_{n\in\mathbb{Z}_+}$ is an $\{\mathcal{F}_n\}$-martingale, then*

$$\mathbb{E}[X_\tau \mid \mathcal{F}_\sigma] = X_{\sigma\wedge\tau} \qquad a.s. \tag{2.42}$$

- *If $\mathbf{X}(\omega)$ is an $\{\mathcal{F}_n\}$-submartingale, then*

$$\mathbb{E}[X_\tau \mid \mathcal{F}_\sigma] \ge X_{\sigma\wedge\tau} \qquad a.s. \tag{2.43}$$

When $\mathbf{X} = (X_n)_{n\in\mathbb{Z}_+}$ is an $\{\mathcal{F}_n\}$-martingale, $\mathbf{X} = (X_n)_{n\in\mathbb{Z}_+}$ and $-\mathbf{X} = (-X_n)_{n\in\mathbb{Z}_+}$ are both $\{\mathcal{F}_n\}$-submartingale. Therefore, if we have (2.43) then since

$$\mathbb{E}[X_\tau \mid \mathcal{F}_\sigma] \ge X_{\sigma\wedge\tau}, \qquad -\mathbb{E}[X_\tau \mid \mathcal{F}_\sigma] \ge -X_{\sigma\wedge\tau} \tag{2.44}$$

we obtain (2.42). Hence it is sufficient to show (2.43). Moreover, since

$$\mathbb{E}[|X_\tau|] = \sum_{n=0}^{N} \mathbb{E}[|X_n|\, \mathbb{1}_{\{\tau=n\}}] \le \sum_{n=0}^{N} \mathbb{E}[|X_n|] < \infty \tag{2.45}$$

we can say that X_τ is measurable.

Proof. We separate the proof into two steps.

Step 1. We first prove that

$$\mathbb{E}[X_\tau \mid \mathcal{F}_k] \ge X_{\tau\wedge k} \quad \text{a.s.}, \qquad (k = 0, 1, \cdots, N-1) \tag{2.46}$$

By the condition $0 \le \tau(\omega) \le N$, for $\Lambda \in \mathcal{F}_{N-1}$,

$$\mathbb{E}[X_\tau\, \mathbb{1}_\Lambda] = \mathbb{E}[X_\tau\, \mathbb{1}_{\Lambda\cap\{\tau=N\}}] + \mathbb{E}[X_\tau\, \mathbb{1}_{\Lambda\cap\{\tau\le N-1\}}] \tag{2.47}$$

We note here that $\{\tau = N\} = \{\tau \le N-1\}^c \in \mathcal{F}_{N-1}$. Therefore, by submartingale property, we have the following inequality

$$\begin{aligned} \mathbb{E}[X_\tau\, \mathbb{1}_{\Lambda\cap\{\tau=N\}}] &= \mathbb{E}[X_N\, \mathbb{1}_{\Lambda\cap\{\tau=N\}}] \\ &\ge \mathbb{E}[X_{N-1}\, \mathbb{1}_{\Lambda\cap\{\tau=N\}}] \\ &= \mathbb{E}[X_{\tau\wedge(N-1)}\, \mathbb{1}_{\Lambda\cap\{\tau=N\}}] \end{aligned} \tag{2.48}$$

Substituting this into (2.47),

$$\mathbb{E}[X_\tau \, \mathbb{1}_\Lambda] \geq \mathbb{E}[X_{\tau\wedge(N-1)} \, \mathbb{1}_\Lambda] \tag{2.49}$$

Since $\mathcal{F}_{\tau\wedge(N-1)} \subset \mathcal{F}_{N-1}$ and $X_{\tau\wedge(N-1)}$ is $\mathcal{F}_{N-1}$-measurable,

$$\mathbb{E}[X_\tau \mid \mathcal{F}_{N-1}] \geq X_{\tau\wedge(N-1)} \quad \text{a.s.} \tag{2.50}$$

By the property of conditional expectation, we obtain

$$\begin{aligned}
\mathbb{E}[X_\tau \mid \mathcal{F}_k] &= \mathbb{E}[\mathbb{E}[X_\tau \mid \mathcal{F}_{N-1}] \mid \mathcal{F}_k] \\
&\geq \mathbb{E}[X_{\tau\wedge(N-1)} \mid \mathcal{F}_k] \\
&= \mathbb{E}[\mathbb{E}[X_{\tau\wedge(N-1)} \mid \mathcal{F}_{N-2}] \mid \mathcal{F}_k] \\
&\geq \mathbb{E}[X_{\tau\wedge(N-2)} \mid \mathcal{F}_k] \\
&= \cdots \\
&\geq X_{\tau\wedge k} \quad \text{a.s.}
\end{aligned} \tag{2.51}$$

Step 2. Since for $\Lambda \in \mathcal{F}_\sigma$, $\Lambda \cap \{\sigma = k\} \in \mathcal{F}_k$, by (2.46),

$$\begin{aligned}
\mathbb{E}[X_\tau \, \mathbb{1}_\Lambda] &= \sum_{k=0}^{\infty} \mathbb{E}[X_\tau \, \mathbb{1}_{\Lambda\cap\{\sigma=k\}}] \\
&\geq \sum_{k=0}^{\infty} \mathbb{E}[X_{\tau\wedge k} \, \mathbb{1}_{\Lambda\cap\{\sigma=k\}}] \\
&= \sum_{k=0}^{\infty} \mathbb{E}[X_{\tau\wedge\sigma} \, \mathbb{1}_{\Lambda\cap\{\sigma=k\}}] \\
&= \mathbb{E}[X_{\tau\wedge\sigma} \, \mathbb{1}_\Lambda]
\end{aligned} \tag{2.52}$$

which concludes (2.43). ■

Remark 2.27. Step 2 utilizes the disjoint decomposition of Ω based on the values of the stopping time σ, effectively patching together the local estimates obtained in Step 1. □

Corollary 2.28. *Let $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ be an $\{\mathcal{F}_n\}$-submartingale and $\tau \leq N$ be an $\{\mathcal{F}_n\}$-stopping time. Then we have*

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_\tau] \leq \mathbb{E}[X_N]. \tag{2.53}$$

Question 2.29. Prove Corollary 2.28.

Remark 2.30. If we take $\sigma = 0$ (which is always a stopping time since $\{0 \leq k\} = \Omega \in \mathcal{F}_k$), then $\mathcal{F}_0$ is the initial σ-algebra. In this case, $X_{\sigma\wedge\tau} = X_0$, and the Optional Sampling Theorem implies:

$$\mathbb{E}[X_\tau \mid \mathcal{F}_0] = X_0 \quad \text{a.s.} \implies \mathbb{E}[X_\tau] = \mathbb{E}[X_0]. \tag{2.54}$$

This simpler form is the most frequently used in applications, such as

- Gambler's Ruin: Finding the probability of hitting a boundary.
- Expected Hitting Time: Calculating $\mathbb{E}[\tau]$ by using martingales like

$$M_n = S_n^2 - n \tag{2.55}$$

□

Question 2.31 (Stopping process)**.** Let $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ be an $\{\mathcal{F}_n\}$-submartingale and $\tau \le N$ be an $\{\mathcal{F}_n\}$-stopping time. we set

$$X_n^\tau := X_{\tau\wedge n}, \qquad \mathbf{X}^\tau := (X_n^\tau)_{n\in\mathbb{Z}_+} \tag{2.56}$$

which is called stopping process. Prove that $\mathbf{X}^\tau$ is an $\{\mathcal{F}_{\tau\wedge n}\}$-submartingale, moreover, it is also an $\{\mathcal{F}_n\}$-submartingale.

Question 2.32. Let $a < 0 < b$ be integers, $S = \{S_n\}_{n\in\mathbb{Z}_+}$ be a symmetric simple random walk starting at 0.We define the stopping time τ as the first time when the symmetric SRW hits a or b, i.e.

$$\tau = \min\{n \ge 0 : S_n = a \text{ or } S_n = b\} \tag{2.57}$$

Compute $\mathbb{E}[\tau]$.

2.6.3 Doob's inequality

Theorem 2.33. *Let $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ be an $\{\mathcal{F}_n\}$-submartingale and λ be a positive constant. Then we have*

$$\lambda\,\mathbb{P}\left(\max_{k\le N} X_k \ge \lambda\right) \le \mathbb{E}\left[X_N\,\mathbb{1}_{\{\max_{k\le N} X_k\ge\lambda\}}\right] \le \mathbb{E}[X_N^+] \tag{2.58}$$

$$\lambda\,\mathbb{P}\left(\min_{k\le N} X_k \le -\lambda\right) \le \mathbb{E}\left[X_N\,\mathbb{1}_{\{\min_{k\le N} X_k\le-\lambda\}}\right] - \mathbb{E}[X_0] \le \mathbb{E}[X_N^+] - \mathbb{E}[X_0] \tag{2.59}$$

$$\lambda\,\mathbb{P}\left(\max_{k\le N} |X_k| \ge \lambda\right) \le 2\mathbb{E}[X_N^+] - \mathbb{E}[X_0] \le 3\max_{k\le N}\mathbb{E}[|X_k|] \tag{2.60}$$

* X_N^+ stands for $\max\{X_N, 0\} = X_N \vee 0$

Proof. First $\tau := \min\{k \ge 0\colon X_k \ge \lambda\}$ is an $\{\mathcal{F}_n\}$-stopping time. By the optional sampling theorem,

$$\mathbb{E}[\mathbb{E}[X_N \mid \mathcal{F}_{\tau\wedge N}]] \ge \mathbb{E}[X_{\tau\wedge N}] = \mathbb{E}[X_\tau\,\mathbb{1}_{\{\tau\le N\}}] + \mathbb{E}[X_N\,\mathbb{1}_{\{\tau\ge N+1\}}] \tag{2.61}$$

Since the left-hand side is

$$LHS = \mathbb{E}[X_N] = \mathbb{E}[X_N\,\mathbb{1}_{\{\tau\le N\}}] + \mathbb{E}[X_N\,\mathbb{1}_{\{\tau\ge N+1\}}], \tag{2.62}$$

we have

$$\mathbb{E}[X_N\,\mathbb{1}_{\{\tau\le N\}}] \ge \mathbb{E}[X_\tau\,\mathbb{1}_{\{\tau\le N\}}] \tag{2.63}$$

On the other hand, by the definition of τ,

$$\{\tau \le N\} = \{\max_{k\le N} X_k \ge \lambda\} \tag{2.64}$$

which implies that $X_\tau \ge \lambda$. Therefore

$$\mathbb{E}[X_N\,\mathbb{1}_{\{\tau\le N\}}] \ge \lambda\,\mathbb{P}(\tau \le N) = \lambda\,\mathbb{P}\left(\max_{k\le N} X_k \ge \lambda\right) \tag{2.65}$$

This proves (2.58). ■

Question 2.34. Prove (2.59) and (2.60).

Example 2.35 (Kolmogorov's inequality)**.** Let ξ_n $(n = 0, 1, \cdots)$ be an i.i.d. random variables with

$$\mathbb{E}[\xi_n] = 0, \qquad \mathbb{V}\mathrm{ar}(\xi_n) = v_n. \tag{2.66}$$

We set

$$\mathcal{F}_n = \sigma(\xi_0, \xi_1, \cdots, \xi_n), \qquad X_n = \sum_{k=0}^{n} \xi_k. \tag{2.67}$$

Then $\mathbf{X} = (X_n)_{n\in\mathbb{Z}_+}$ is an $\{\mathcal{F}_n\}$-martingale and $(X_n^2)_{n\in\mathbb{Z}_+}$ is an $\{\mathcal{F}_n\}$-submartingale. Therefore, we have

$$\mathbb{P}\left(\max_{k\le N}\left|\sum_{i=0}^{k}\xi_i\right| \ge \lambda\right) \le \frac{1}{\lambda^2}\,\mathbb{E}[|X_N|^2] = \sum_{i=0}^{N}\frac{v_i}{\lambda^2} \tag{2.68}$$

□

Next we show the L^p-version inequality for $\{\mathcal{F}_n\}$-martingales.

Theorem 2.36. *Let $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ be L^p and an $\{\mathcal{F}_n\}$-martingale. Then we have*

$$(1) \quad \lambda\,\mathbb{P}\left(\max_{k\le N}|X_k| \ge \lambda\right) \le \mathbb{E}[|X_N|] \tag{2.69}$$

$$(2) \quad \left(\mathbb{E}\left[\max_{k\le N}|X_k|^p\right]\right)^{\frac{1}{p}} \le \frac{p}{p-1}\left(\mathbb{E}[|X_N|^p]\right)^{\frac{1}{p}} \qquad p > 1 \tag{2.70}$$

Proof. Since $(|X_n|)_{n\in\mathbb{Z}_+}$ is an $\{\mathcal{F}_n\}$-submartingale, by the inequality (2.58) in Theorem 2.33, we can obtain (1).

Set $Y = \max_{k\le N}|X_k|$. Then Y is also L^p due to $Y \le \sum_{k=0}^{N}|X_k|$, and

$$\mathbb{E}[Y^p] = \mathbb{E}\left[\int_0^Y p\,x^{p-1}\,dx\right] = \mathbb{E}\left[\int_0^\infty \mathbb{1}_{[0,Y]}(x)\,px^{p-1}\,dx\right] = \int_0^\infty \mathbb{P}(Y \ge x)\,px^{p-1}\,dx \tag{2.71}$$

On the other hand, by (2.58), we have

$$\mathbb{P}(Y \ge x) \le \frac{1}{x}\,\mathbb{E}[|X_N|\,\mathbb{1}_{\{Y\ge N\}}] \tag{2.72}$$

Now we substitute it into (2.71) and obtain

$$\mathbb{E}[Y^p] \le \mathbb{E}\left[|X_N|\int_0^Y px^{p-2}\,dx\right] = \mathbb{E}\left[|X_N|\,\frac{p}{p-1}Y^{p-1}\right] \le \frac{p}{p-1}\left(\mathbb{E}[|X_N|^p]\right)^{\frac{1}{p}}\left(\mathbb{E}[Y^p]\right)^{\frac{p-1}{p}} \tag{2.73}$$

which yields (2.70). ■

$$\begin{aligned}\mathbb{E}[Y^p] &= \int_0^\infty px^{p-1}\mathbb{P}(Y \ge x)dx \\ &\le \int_0^\infty px^{p-1}\left(\frac{1}{x}\mathbb{E}[|X_N|\mathbb{1}_{Y\ge x}]\right)dx \\ &= \mathbb{E}\left[|X_N|\int_0^Y px^{p-2}dx\right] = \frac{p}{p-1}\mathbb{E}[|X_N|Y^{p-1}]\end{aligned} \tag{2.74}$$

2.6.4 Convergence theorem

The goal of this section is to prove the almost sure convergence of $\{\mathcal{F}_n\}$-submartingale, which is called *martingale convergence theorem*. The theorem is corresponding to the convergence of monotone and bounded sequences.

Doob's upcrossing lemma The worst scenario that a martingale does not converge is that its value fluctuates so long and never settle down. We count the number of upcrossings of interval $[a, b]$ by the graph $y = X_n$ to estimate how the martingale fluctuates. Set

$$\sigma_1 = \min\{n \geq 0 \colon X_n \leq a\}, \qquad \tau_1 = \min\{n \geq \sigma_1 + 1 \colon X_n \geq b\} \tag{2.75}$$

σ_1 is the first time X_n crossing a from above and τ_1 is the first time X_n crossing b from below after σ_1. Repeatedly we define

$$\sigma_k = \min\{n \geq \tau_{k-1} + 1 \colon X_n \leq a\}, \qquad \tau_k = \min\{n \geq \sigma_k + 1 \colon X_n \geq b\} \tag{2.76}$$

where $\min \varnothing = \infty$ for convention. Then (σ_k) and (τ_k) are $\{\mathcal{F}_n\}$-stopping times. The number of upcrossings of interval $[a, b]$ until time N can be expressed by

$$U_N(a, b) := \sum_{k=1}^{\infty} \mathbb{1}_{[0,N]}(\tau_k), \tag{2.77}$$

where

$$\mathbb{1}_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases} \tag{2.78}$$

Since the summation has the meaning only for k such that $\tau_k \leq N$, it is actually a finite sum. If $U_\infty(a, b) < \infty$, then the fluctuation on the interval $[a, b]$ will be settled down.

Theorem 2.37 (Doob's upcrossing lemma)**.**

$$\mathbb{E}[U_N(a, b)] \leq \frac{1}{b - a} \mathbb{E}[(X_N - a)^+ - (X_0 - a)^+]. \tag{2.79}$$

Proof. We set $Y_n = (X_n - a)^+$. Then $\mathbf{Y} = (Y_n)_{n \in \mathbb{Z}_+}$ is an $\{\mathcal{F}_n\}$-submartingale, and $U_N(a, b)$ describes the number of upcrossings that $\mathbf{Y}$ crosses the interval $[0, b - a]$ upward. Moreover, $Y_{\sigma_k} = 0$ and $Y_{\tau_k} \geq b - a$. We define the following variables:

$$\begin{aligned} \xi &= \sum_{k=1}^{N} (Y_{\sigma_k \wedge N} - Y_{\tau_k \wedge N}), \\ \eta &= \sum_{k=1}^{N} (Y_{\tau_k \wedge N} - Y_{\sigma_k \wedge N}) \end{aligned} \tag{2.80}$$

Then since $\tau_\ell \leq N < \tau_{\ell+1}$ where $\tau_0 = 0$ and $U_N(a, b) = \ell$, it is enough to consider the following two cases:

1. $\sigma_\ell < \tau_\ell \leq N < \sigma_{\ell+1}$
2. $\tau_\ell < \sigma_{\ell+1} \leq N < \tau_{\ell+1}$

The case **1.** When $\ell = 0$, $\xi = \eta = 0$ since $\sigma_k, \tau_k > N$ if $k \geq 1$. For $\ell \geq 1$, since $\sigma_\wedge N = \tau_k \wedge N = N$ if $k \geq \ell + 1$,

$$\begin{aligned} \xi &= Y_N - Y_{\tau_\ell} + \sum_{k=1}^{\ell-1} (Y_{\sigma_{k+1}} - Y_{\tau_k}) = Y_N - \sum_{k=1}^{\ell} Y_{\tau_k} \\ \eta &= \sum_{k=1}^{\ell} (Y_{\tau_k} - Y_{\sigma_k}) = \sum_{k=1}^{\ell} Y_{\tau_k} \geq \ell(b - a) \end{aligned} \tag{2.81}$$

because $Y_{\sigma_k} = 0$.

The case **2.** When $\ell = 0$, $\xi = 0$ and $\eta = Y_N - Y_{\sigma_1} = Y_N \geq 0$. For $\ell \geq 1$, similarly to the case 1,

$$\begin{aligned} \xi &= \sum_{k=1}^{\ell} (Y_{\sigma_{k+1}} - Y_{\tau_k}) = -\sum_{k=1}^{\ell} Y_{\tau_k} \\ \eta &= Y_N - Y_{\ell+1} + \sum_{k=1}^{\ell} (Y_{\tau_k} - Y_{\sigma_k}) = Y_N + \sum_{k=1}^{\ell} Y_{\tau_k} \end{aligned} \tag{2.82}$$

For $\ell \geq 1$, $Y_{\sigma_1 \wedge N} = Y_{\sigma_1} = 0$ since $\sigma_1 \wedge N = \sigma_1$. For $\ell = 0$,

- in the case **1.**: since $\xi + \eta = 0$ and $\sigma_1 \wedge N = N$,

$$Y_N - Y_{\sigma_1 \wedge N} = 0 \tag{2.83}$$

- in the case **2.**: since $\xi + \eta = Y_N$ and $\sigma_1 \wedge N = \sigma_1$,

$$Y_N - Y_{\sigma_1 \wedge N} = 0 \tag{2.84}$$

Therefore, in both cases, we have

$$\xi + \eta = Y_N - Y_{\sigma_1 \wedge N}. \tag{2.85}$$

Since by the submartingale property of $\mathbf{Y}$,

$$\mathbb{E}[\xi] \geq 0, \qquad \mathbb{E}[Y_{\sigma_1 \wedge N} - Y_0] \geq 0, \tag{2.86}$$

we have

$$\mathbb{E}[Y_N] - \mathbb{E}[Y_0] = \mathbb{E}[\xi] + \mathbb{E}[\eta] + \mathbb{E}[Y_{\sigma_1 \wedge N} - Y_0] \geq \mathbb{E}[\eta] \tag{2.87}$$

Since $\eta \geq U_N(a,b)(b-a)$,

$$\mathbb{E}[U_N(a,b)(b-a)] \leq \mathbb{E}[\eta] \leq \mathbb{E}[Y_N] - \mathbb{E}[Y_0] \tag{2.88}$$

Therefore, by the definition of Y_n, we obtain (2.79). ■

Since $U_N(a,b)$ is increasing in N, we obtain the following corollary by the monotone convergence theorem.

Corollary 2.38. *If* $\sup_n \mathbb{E}[X_n^+] < \infty$*, then we have*

$$\sup_{N \geq 1} \mathbb{E}[U_N(a,b)] = \mathbb{E}\left[\lim_{n \to \infty} U_N(a,b)\right] \leq \frac{1}{b-a}\left(\sup_n \mathbb{E}[X_n^+] + |a|\right) \tag{2.89}$$

We note here that

$$\mathbb{E}[Y_N] = \mathbb{E}[(X_N - a)^+] \leq \mathbb{E}[X_N^+ + a^-] = \mathbb{E}[X_N^+ + |a| \mathbb{1}_{\{a<0\}}] \tag{2.90}$$

Remark 2.39 (Intuition behind crossing lemma)**.** The fundamental reason we study the number of upcrossings $U_N(a,b)$ is to eliminate the possibility of limitless oscillation. In deterministic analysis (in calculus), a bounded sequence converges if it does not oscillate infinitely. For a submartingale, although it increases on average, the individual sample paths could still oscillate widely. The upcrossing lemma provides a crutial connection between the submartingale property and pathwise stability, It tells us that the probability of a path frequently jumping from below a to above b is strictly bounded by its total expectation growth. Thus, infinite oscillation does not occur almost surely. □

Martingale convergence theorem Thanks to Doob's upcrossing lemma, we can see submartingales increase on average. Combining with the boundedness, we can have similar theorem to that for bounded monotone sequences. The following convergence theorem was proven for submartingales by Doob.

Theorem 2.40 (Martingale convergence theorem)**.** *Let $(X_n)_{n\in\mathbb{Z}_+}$ be an $\{\mathcal{F}_n\}$-submartingale satisfying $\sup_n \mathbb{E}[X_n^+] < \infty$. Then X_n converges almost surely and the limit X_∞ is integrable.*

Proof. we set $m = \sup_n \mathbb{E}[X_n^+]$. We separate the proof into three steps:

(1). Eliminate the divergence

(2). Existence of the a.s. limit: $\mathbb{P}\left(\limsup_{n\to\infty} X_n > \liminf_{n\to\infty} X_n\right) = 0$

(3). Integrability of the limit

(1). By the inequality of submartingale (2.58),

$$\mathbb{P}\left(\max_{k\le N} X_k \ge \lambda\right) \le \frac{1}{\lambda}\mathbb{E}[X_N^+] \le \frac{m}{\lambda} \tag{2.91}$$

Since the event $\{\max_{k\le N} X_k \ge \lambda\}$ approaches monotonically to the event $\{\sup_k X_k \ge \lambda\}$ as $N \to \infty$, by the monotone convergence theorem[1)], we obtain

$$\mathbb{P}\left(\sup_k X_k \ge \lambda\right) \le \frac{m}{\lambda} \tag{2.93}$$

and taking $\lambda \to \infty$, we obtain by the continuity of probability that

$$\mathbb{P}\left(\sup_n X_n = \infty\right) = 0. \tag{2.94}$$

Similarly, since by (2.59) in Theorem 2.33,

$$\mathbb{P}\left(\min_{k\le N} X_k \le -\lambda\right) \le \frac{1}{\lambda}\left(\mathbb{E}[X_N^+] - \mathbb{E}[X_0]\right) \le \frac{1}{\lambda}(m + \mathbb{E}[|X_0|]) \tag{2.95}$$

we obtain as $\lambda \to \infty$

$$\mathbb{P}\left(\inf_n X_n = -\infty\right) = 0 \tag{2.96}$$

(2). Now we have

$$\begin{aligned}\left\{\limsup_{n\to\infty} X_n > \liminf_{n\to\infty} X_n\right\} &= \bigcup_{a,b\in\mathbb{Q},\, a<b} \left\{\limsup_{n\to\infty} X_n > b > a > \liminf_{n\to\infty} X_n\right\} \\ &\subset \bigcup_{a,b\in\mathbb{Q},\, a<b} \left\{\lim_{N\to\infty} U_N(a,b) = \infty\right\}\end{aligned} \tag{2.97}$$

Since by Corollary 2.38, $\lim_{N\to\infty} U_N(a,b)$ is measurable and almost surely bounded,

$$\mathbb{P}\left(\lim_{N\to\infty} U_N(a,b) = \infty\right) = 0 \tag{2.98}$$

Therefore, we obtain

$$\mathbb{P}\left(\limsup_{n\to\infty} X_n > \liminf_{n\to\infty} X_n\right) = 0 \tag{2.99}$$

[1)] Here we consider

$$\mathbb{1}_{\{\max_{k\le N} X_k \ge \lambda\}} \uparrow \mathbb{1}_{\{\sup_k X_k \ge \lambda\}} \qquad N \to \infty \tag{2.92}$$

(3). By (2.99), X_n converges almost surely. We set $X_\infty := \lim_{n\to\infty} X_n$. Then, by Lebesgue–Fatou lemma,

$$\mathbb{E}[|X_\infty|] \le \liminf_{n\to\infty} \mathbb{E}[|X_n|] \le \liminf_{n\to\infty} \mathbb{E}[(2X_n^+ - X_n)] \tag{2.100}$$

Since by the submartingale property of X_n, we have $\mathbb{E}[X_0] \le \mathbb{E}[X_n]$,

$$RHS = \liminf_{n\to\infty} \mathbb{E}[(2X_n^+ - X_n)] \le 2m - \mathbb{E}[X_0] \tag{2.101}$$

which implies that X_∞ is integrable. ■

Corollary 2.41. *Let $\mathbf{X}(\omega) = (X_n)_{n\in\mathbb{Z}_+}$ be a nonnegative, $\{\mathcal{F}_n\}$-martingale. Then X_n converges almost surely.*

* We remark here that we implicitly assumed that X_0 is integrable.

Proof. We first note that $X_n^+ = X_n$. Thus

$$\mathbb{E}[X_n^+] = \mathbb{E}[X_0] \tag{2.102}$$

which implies the condition $\sup_n \mathbb{E}[X_n^+] < \infty$. Therefore, we can apply Theorem 2.40. ■

The following theorem states that as our "information" (the σ-algebra $\mathcal{F}_n$) grows, our best estimate of random variable X, which is $\mathbb{E}[X \mid \mathcal{F}_n]$) gradually approaches the "ultimate" estimate based on all information available in the limit. If X is $\mathcal{F}_\infty$-measurable, then $X_n \to X$ a.s., meaning that we eventually know X perfectly.

Theorem 2.42. *Let X be a square integrable random variable on $(\Omega, \mathcal{F}, \{\mathcal{F}_n\}, \mathbb{P})$, $\mathcal{F}_\infty$ be a smallest σ-algebra containing $\mathcal{F}_0, \mathcal{F}_1, \cdots$, i.e.*

$$\mathcal{F}_\infty = \sigma\left(\bigcup_{n\ge 0} \mathcal{F}_n\right). \tag{2.103}$$

Then as $n \to \infty$ we have

$$\mathbb{E}[X \mid \mathcal{F}_n] \to \mathbb{E}[X \mid \mathcal{F}_\infty] \tag{2.104}$$

almost surely and in $L^1(\Omega)$.

Proof. We set $X_n = \mathbb{E}[X \mid \mathcal{F}_n]$. Since $\mathbf{X} = (X_n)_{n\in\mathbb{Z}_+}$ is an $\{\mathcal{F}_n\}$-martingale and $\mathbb{E}[|X_n|] \le \mathbb{E}[X]$, by Theorem 2.40, X_n converges almost surely.

Moreover, since

$$\mathbb{E}[X_n^2] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_n]^2] \le \mathbb{E}[\mathbb{E}[X^2 \mid \mathcal{F}_n]] = \mathbb{E}[X^2] \tag{2.105}$$

$\mathbf{X}$ is uniformly integrable. Thus X_n converges to X_∞ in L^1. Since X_n is $\mathcal{F}_n$-measurable, it is also $\mathcal{F}_\infty$-measurable and then X_∞ is also $\mathcal{F}_\infty$-measurable.

For arbitrary $\ell \in \mathbb{Z}_+$ and $\Lambda \in \mathcal{F}_\ell$,

$$\mathbb{E}[X_n \mathbb{1}_\Lambda] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_n] \mathbb{1}_\Lambda] = \mathbb{E}[X \mathbb{1}_\Lambda] \qquad n \ge \ell \tag{2.106}$$

Since X_n converges to X_∞ in L^1, we have

$$\mathbb{E}[X_\infty \mathbb{1}_\Lambda] = \lim_{n\to\infty} \mathbb{E}[X_n \mathbb{1}_\Lambda] = \mathbb{E}[X \mathbb{1}_\Lambda] \tag{2.107}$$

The above is held for $\Lambda \in \bigcup_\ell \mathcal{F}_\ell$. Since $\bigcup_\ell \mathcal{F}_\ell$ can generate $\mathcal{F}_\infty$, (2.107) holds for $\Lambda \in \mathcal{F}_\infty$, which yields

$$X_\infty = \mathbb{E}[X \mid \mathcal{F}_\infty]. \tag{2.108}$$

■

Remark 2.43. While almost sure convergence tells us about the behavior of individual paths, L^1-convergence ensures that no "mass" escapes to infinity. Since $X_n = \mathbb{E}[X \mid \mathcal{F}_n]$ is formed by averaging a single integrable X, the sequence $\{X_n\}$ cannot fluctuate widely enough to lose its integrability in the limit. □

Question 2.44 (Kolmogorov's LLN)**.** Let $(\xi_n)_{n=1,2,\cdots}$ be a sequence of square integrable i.i.d. random variables with

$$\mathbb{E}[\xi_n] = 0, \qquad \mathbb{V}\mathrm{ar}(\xi_n) = v < \infty. \tag{2.109}$$

Then prove via the martingale convergence theorem that

$$\frac{1}{n}\sum_{k=1}^{n} \xi_k \to 0 \qquad \text{a.s.} \tag{2.110}$$

as $n \to \infty$.

Downward martingale convergence theorem While the previous theorem (Theorem 2.40) dealt with the accumulation of information ($\mathcal{F}_n \uparrow \mathcal{F}_\infty$), we now consider the case where the information is being "refined" or "summarized" as n increases, represented by a decreasing sequence of σ-algebras.

Theorem 2.45 (Downward convergence theorem)**.** *Let $(\mathcal{G}_n)_{n\in\mathbb{Z}_+}$ be a decreasing sequence of σ-algebras, i.e., $\mathcal{G}_0 \supset \mathcal{G}_1 \supset \mathcal{G}_2 \supset \cdots$, and let*

$$\mathcal{G}_\infty = \bigcap_{n=0}^{\infty} \mathcal{G}_n. \tag{2.111}$$

For any integrable random variable $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$, the sequence $X_n := \mathbb{E}[X \mid \mathcal{G}_n]$ converges almost surely and in $L^1(\Omega)$ to $\mathbb{E}[X \mid \mathcal{G}_\infty]$ as $n \to \infty$.

Remark 2.46. Note that for an increasing sequence of σ-algebras $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$, the simple union $\bigcup_{n\geq 0} \mathcal{F}_n$ is generally not a σ-algebra. For instance, if $\Omega = \mathbb{N}$ and $\mathcal{F}_n = \sigma(\{1\}, \ldots, \{n\})$, the set of all even numbers is not contained in $\bigcup_{n\geq 0} \mathcal{F}_n$, although it belongs to $\sigma(\bigcup_{n\geq 0} \mathcal{F}_n)$. Thus, the operator $\sigma(\cdot)$ is essential for the definition of $\mathcal{F}_\infty$. In contrast, for a decreasing sequence $(\mathcal{G}_n)_{n\in\mathbb{Z}_+}$, the intersection $\mathcal{G}_\infty = \bigcap_{n=0}^{\infty} \mathcal{G}_n$ is automatically a σ-algebra due to the closure property of σ-algebras under countable intersections. Therefore, no additional generation is required for the limit of a decreasing sequence. □

Proof. The sequence (X_n) is called a *backward martingale* (or *reverse martingale*). First, we note that $\{X_n\}_{n\geq 0}$ is **uniformly integrable** because it is a collection of conditional expectations of a single integrable random variable X.

For the almost sure convergence, we apply Doob's upcrossing lemma in "reverse time". For $a < b$, let $U_N(a, b)$ be the number of upcrossings of $[a, b]$ by the finite sequence

$$(X_N, X_{N-1}, \ldots, X_0). \tag{2.112}$$

By (2.79), we have

$$\mathbb{E}[U_N(a, b)] \leq \frac{\mathbb{E}[(X_0 - a)^+]}{b - a}. \tag{2.113}$$

As $N \to \infty$, $U_N(a, b)$ increases monotonically to $U_\infty(a, b)$. By the monotone convergence theorem, $\mathbb{E}[U_\infty(a, b)] < \infty$, which implies $U_\infty(a, b) < \infty$ a.s. Since this holds for all $a, b \in \mathbb{Q}$, X_n converges almost surely to some X_∞.

The L^1-convergence follows directly from the uniform integrability. Finally, since X_n is $\mathcal{G}_n$-measurable, X_∞ is $\mathcal{G}_\infty$-measurable. The identity $X_\infty = \mathbb{E}[X \mid \mathcal{G}_\infty]$ is obtained by the same argument as in Theorem 2.42. ■

Remark 2.47. Qualitatively, this theorem describes the behavior of our estimates as we "forget" specific details and retain only the "tail" or "invariant" information. Note that unlike the upward case, we do not need the additional condition $\sup \mathbb{E}[X_n^+] < \infty$, as the structure $X_n = \mathbb{E}[X \mid \mathcal{G}_n]$ automatically provides L^1-boundedness and uniform integrability. □

Question 2.48 (SLLN via Backward Martingale)**.** Let $(\xi_n)_{n=1,2,\dots}$ be a sequence of i.i.d. L^1 random variables with

$$\mathbb{E}[\xi_1] = \mu. \tag{2.114}$$

By considering the decreasing σ-algebras $\mathcal{G}_n = \sigma(S_n, S_{n+1}, S_{n+2}, \dots)$, prove that as $n \to \infty$

$$\frac{1}{n}\sum_{k=1}^{n} \xi_k \to 0 \qquad \text{a.s.} \tag{2.115}$$

A Zero-One Law and Strong Law of Large Numbers (not completed)

Comparing Question 2.44 and 2.48, the assumption in Question 2.48 is more general. Less conditioned makes the analysis harder, and in this case we need a technical argument (0-1 law).

Definition A.1 (Exchangeable sequence)**.** A sequence of random variables $(\xi_n)_{n=1,2,\dots}$ is said to be **exchangeable** if for every $n \geq 1$ and every permutation π of $\{1, \dots, n\}$, the joint distribution of $(\xi_1, \dots, \xi_n)$ is the same as that of $(\xi_{\pi(1)}, \dots, \xi_{\pi(n)})$. That is, for any Borel sets $B_1, \dots, B_n \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(\xi_1 \in B_1, \dots, \xi_n \in B_n) = \mathbb{P}(\xi_{\pi(1)} \in B_1, \dots, \xi_{\pi(n)} \in B_n). \tag{A.116}$$

Definition A.2 (Exchangeable σ-algebra)**.** Let $(\xi_n)_{n=1,2,\dots}$ be a sequence of random variables. An event $\Lambda \in \sigma(\xi_1, \xi_2, \dots)$ is called **exchangeable** if it is invariant under any finite permutation of the indices. Specifically, let π be a permutation of $\mathbb{Z}_+$ that fixes all but finitely many elements. If we define $\xi_n^\pi = \xi_{\pi(n)}$, then Λ is exchangeable if

$$\mathbb{1}_\Lambda(\xi_1, \xi_2, \dots) = \mathbb{1}_\Lambda(\xi_1^\pi, \xi_2^\pi, \dots) \quad \text{a.s.} \tag{A.117}$$

The collection of all such events forms a σ-algebra $\mathcal{E}$, called the **exchangeable σ-algebra**.

Theorem A.3 (Hewitt–Savage 0-1 law)**.** *Let $(\xi_n)_{n=1}^\infty$ be a sequence of i.i.d. random variables. Let $\mathcal{E}$ be the exchangeable σ-algebra, defined as:*

$$\mathcal{E} = \{\Lambda \in \mathcal{F} : \Lambda \text{ is invariant under any finite permutation of } (\xi_n)\}. \tag{A.118}$$

Then, for any $\Lambda \in \mathcal{E}$, we have $\mathbb{P}(\Lambda) \in \{0, 1\}$. Consequently, any $\mathcal{E}$-measurable random variable is almost surely constant.

The convergence of the sample mean $\frac{S_n}{n}$ can be elegantly explained by Theorem A.3 as follows:

(1) **Invariance:** The limit

$$X_\infty = \lim_{n\to\infty} \frac{S_n}{n} \tag{A.119}$$

(if it exists) is invariant under any finite permutation of the sequence (ξ_n). For example, swapping ξ_1 and ξ_2 does not change the value of $\frac{S_n}{n}$ for $n \geq 2$, and thus leaves the limit unchanged.

(2) **Measurability:** This implies X_∞ is measurable with respect to the exchangeable σ-algebra $\mathcal{G}_\infty \subset \mathcal{E}$.

(3) **Conclusion:** By Theorem A.3, X_∞ must be a constant a.s. Since we already know $\mathbb{E}[X_\infty] = \mathbb{E}[\xi_1] = \mu$ from the L^1-convergence of the backward martingale, we conclude $X_\infty = \mu$ a.s.

Note that this is stronger than Kolmogorov's 0-1 law, which only applies to *tail events* (events independent of any finite initial segment). The exchangeable σ-algebra $\mathcal{E}$ contains the tail σ-algebra, making Hewitt–Savage the natural tool for symmetric statistics like $\frac{S_n}{n}$.

Definition A.4 (Tail σ-algebra)**.** Let $(X_n)_{n=1,2,\cdots}$ be a sequence of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For each $n \geq 1$, define the σ-algebra generated by the sequence starting from index n:

$$\mathcal{T}_n = \sigma(X_n, X_{n+1}, X_{n+2}, \dots). \tag{A.120}$$

The **tail σ-algebra** $\mathcal{T}$ is defined as the intersection of these decreasing σ-algebras:

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \mathcal{T}_n = \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \dots). \tag{A.121}$$

An event $\Lambda \in \mathcal{T}$ is called a **tail event**.

Remark A.5. A tail event Λ is "independent of the past" in the sense that for any fixed N, Λ does not depend on the values of $(X_1, \dots, X_N)$. Typical examples of tail events include:

- $\{\omega : \sum_{n=1}^{\infty} X_n(\omega) \text{ converges}\}$,
- $\{\omega : \limsup_{n\to\infty} X_n(\omega) > c\}$,
- $\{\omega : \lim_{n\to\infty} \frac{S_n}{n} = \mu\}$.

Theorem A.6 (Kolmogorov's zero-one law)**.** *Let $(\xi_n)_{n=1,2,\cdots}$ be a sequence of independent random variables. Let $\mathcal{T}$ be the **tail σ-algebra**, defined as:*

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \sigma(\xi_n, \xi_{n+1}, \xi_{n+2}, \dots). \tag{A.122}$$

Then, for any $\Lambda \in \mathcal{T}$, we have $\mathbb{P}(\Lambda) \in \{0, 1\}$.

Remark A.7 (Comparison: Kolmogorov vs Hewitt–Savage)**.** It is essential to understand the difference between the tail σ-algebra $\mathcal{T}$ and the exchangeable σ-algebra $\mathcal{E}$:

- **Kolmogorov's $\mathcal{T}$:** Contains events that do not depend on any *finite initial segment* of the sequence. (e.g., $\{\sum \xi_n \text{ converges}\}$).
- **Hewitt–Savage's $\mathcal{E}$:** Contains events that are invariant under any *finite permutation* of indices. (e.g., $\{\limsup_{n\to\infty} \frac{S_n}{n} > c\}$).

Under the i.i.d. assumption, we have the following inclusion of σ-algebras:

$$\mathcal{T} \subset \mathcal{E}. \tag{A.123}$$

Therefore, **Hewitt–Savage is a stronger result** than Kolmogorov's in the i.i.d. setting, as it shows that a larger class of events (exchangeable events) is trivial.